Data Communications and Networking

Online Quiz Chapter 4

T2.2023-2024

1. **Compute the signal-to-quantization noise power ratio (S/N) for 4-bit binary PCM with a sinusoidal analog signal. for n = 8.**
    a. **49.76**
    b. **37.76**
    c. **25.76**
2. **The number of bits per sample used in a standard telephony voice transmission system is 8. Calculate the number of quantized levels in PCM.**
    a. **16**
    b. **8**
    c. **256**
3. **Determine Nyquist interval for an analog signal represented by s(t) = 5 cos (100 π t) + 10 cos(2000 π t).**
    a. **20ms**
    b. **1ms**
    c. **0.5ms**
4. **Telephone companies digitize voice by assuming a maximum frequency of 4KHz.The sampling rate is equal to.**
    a. **2000 sample/s**
    b. **4000 sample/s**
    c. **8000 sample/s**
5. **Asynchronous transmission means.**
    a. **asynchronous at bit level**
    b. **asynchronous at byte level**
    c. **asynchronous at symbol level**

ليث عمرو

Data Communications and Networking

Online Quiz Chapter 4

T2.2023-2024

6. **Companding means.**
    a. **reducing instantaneous voltage amplitude for large values**
    b. **Increasing instantaneous voltage amplitude for small values**
    c. **reducing instantaneous voltage amplitude for small values**
7. **For maximum audio input frequency, determine the minimum sampling.**
    a. **fs ≥ 4 kHz**
    b. **fs ≥ 8 kHz**
    c. **fs ≤ 8 kHz**
8. **Determine the PCM transmission data rate for a single-channel PCM system employing a sample rate of 8000 samples per second and an 8-bit compressed PCM code.**
    a. **1000 kbps**
    b. **64 kbps**
    c. **64 Mbps**
9. **Consider an analog signal given by s(t) = 2 cos (2 π 100t) cos (2 π 10t) is sampled at the rate of 250 samples per seconds. Determine the maximum frequency component present in the signal.**
    a. **220 Hz**
    b. **90 Hz**
    c. **110 Hz**
10. **for short distance usually we use.**
    a. **parallel transmission**
    b. **series transmission**
    c. **synchronous transmission**

ليث عمرو